

Jeffrey A. Young

AI Software Architect | Enterprise AI Consultant | Data Scientist

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Executive Profile

AI software architect, data scientist, and security researcher with more than ten years of experience designing AI, machine learning, cloud analytics, and decision-support solutions. Experienced evaluating emerging AI technologies, translating organizational requirements into secure reference architectures, and developing AI agents, Retrieval-Augmented Generation (RAG) workflows, proof-of-concept criteria, executive dashboards, and adoption KPIs. Brings enterprise experience with Microsoft Azure, AWS, Python, SQL, Power BI, Tableau, data lakes, predictive analytics, and AI security research. Published author at USENIX Security and ACM CCS with a strong record of technical leadership, stakeholder communication, and applied AI innovation.

AI Transformation & Enterprise Architecture

AI Strategy	Enterprise AI roadmaps, use-case prioritization, operating models, adoption planning, and business-value alignment.
AI Governance	Review gates, responsible AI controls, security and privacy assessment, vendor evaluation, risk documentation, and approval criteria.
Architecture	Secure reference architectures, reusable implementation patterns, cloud integration, data pipelines, APIs, and scalable AI services.
Generative AI	AI agents, RAG, ChatGPT, Microsoft Copilot, prompt engineering, conversational systems, knowledge retrieval, and workflow automation.
PoC Leadership	Feasibility criteria, success metrics, KPI definition, pilot design, evaluation plans, transition-to-production criteria, and ROI measurement.
Executive Analytics	AI adoption dashboards, operational KPIs, Power BI, Tableau, visual reporting, and leadership-facing decision support.
Cloud Platforms	Microsoft Azure, Azure Data Lake, Azure AI integration, AWS services, cloud analytics, and hybrid solution design.

Professional Experience

Bosch USA

Data Scientist

2022–2025

United States

- Designed and deployed data-driven solutions across AI, machine learning, predictive analytics, and IoT, with emphasis on predictive maintenance, workflow optimization, and scalable enterprise applications.
- Built and managed Azure data-lake solutions and relational data systems supporting enterprise analytics, reporting, and operational decision-making.
- Developed Python, SQL, R, and C# analytical pipelines and applications for manufacturing and business use cases.
- Created Power BI and Tableau dashboards that translated technical results into operational KPIs and leadership-ready insights.
- Evaluated data, platform, and implementation requirements for proof-of-concept initiatives and production deployment.
- Collaborated with engineering, operations, and business stakeholders to identify high-value use cases and align solutions with measurable outcomes.
- Applied image processing, data engineering, GitHub-based development, and LeanIX-supported workflow and architecture practices.

Clemson University

AI Security Researcher / Graduate Research Assistant

2019–Present

Clemson, SC

- Research voice-assistant platforms, AI application ecosystems, policy compliance, privacy, and automated security evaluation.
- Designed large-scale systems for detecting policy violations and risky behaviors in third-party AI applications using NLP, review analysis, machine learning, and automated testing.
- Developed predictive and sequence models using RNNs, DNNs, Markov models, scikit-learn, PyTorch, and TensorFlow.
- Led collaborative development of a fuzzing and automated testing platform for voice-assistant applications.
- Published peer-reviewed AI and security research at USENIX Security, ACM CCS, IEEE venues, ICCS, and CSCI.
- Communicated complex AI, security, and privacy findings to academic, technical, and interdisciplinary audiences.

- Served as instructor for CPSC 4240/6240, System Administration and Security.
- Taught secure systems administration, operating-system concepts, infrastructure practices, and applied security.
- Developed technical learning materials, assessments, labs, and explanations for undergraduate and graduate students.

Independent AI Consulting & Product Development

Founder / AI Consultant

Current

Remote

- Design AI-powered business tools, intelligent assistants, workflow automation systems, and decision-support applications for consulting, education, research, and digital products.
- Translate stakeholder needs into implementation plans, system requirements, architecture choices, data flows, success measures, and deployment roadmaps.
- Build conversational assistants and knowledge-based systems using AI-agent and RAG patterns.
- Develop AI strategy materials, implementation recommendations, executive reporting, and data-driven business plans.
- Evaluate AI platforms and product capabilities for fit, risk, usability, extensibility, and organizational adoption.

Selected Enterprise AI Engagements

- **AI Business Planning Platform** — Designed an AI-assisted consulting workflow that gathers client inputs and produces structured, data-driven implementation roadmaps, automation opportunities, analytics recommendations, and executive guidance.
- **Enterprise Knowledge Assistants** — Built conversational assistants that combine curated organizational knowledge, retrieval workflows, prompt design, and user-facing interfaces to support research, customer engagement, and decision-making.
- **AI Education and Research Platform** — Created AI-supported educational resources, technical tutorials, research summaries, and interactive assistants for data science, machine learning, AI security, and applied analytics.
- **Digital Product AI Integration** — Designed and implemented AI-powered assistants, content workflows, recommendation concepts, search experiences, and user-facing automation for web-based products and publishing platforms.
- **Predictive Maintenance and Operations Analytics** — Developed data pipelines, models, dashboards, and operational analytics for equipment, manufacturing, and workflow optimization use cases.

Technical Leadership

- Define technical approaches that balance organizational value, security, privacy, scalability, usability, and implementation cost.
- Establish proof-of-concept objectives, acceptance criteria, baseline measures, KPIs, and transition-to-production requirements.
- Explain architecture and AI risk to technical teams, executives, educators, and nontechnical stakeholders.
- Lead interdisciplinary work involving machine learning, security, software development, data engineering, analytics, and visualization.
- Evaluate products and technical approaches through structured comparison of capabilities, integration requirements, limitations, and risk.

Technical Stack

Generative AI	AI agents, Retrieval-Augmented Generation, ChatGPT, Microsoft Copilot, prompt engineering, conversational AI, semantic retrieval, knowledge assistants
AI / Machine Learning	Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, SciPy, NLP, deep learning, predictive modeling, feature engineering
Cloud & Data	Microsoft Azure, Azure Data Lake, AWS, SQL, MySQL, relational design, data lakes, data engineering, cloud analytics
Programming	Python, SQL, Java, JavaScript, C, C++, C#, R, HTML5
Visualization	Power BI, Tableau, Plotly, Matplotlib, R Shiny, executive dashboards, KPI reporting
Engineering Tools	Linux, Git, GitHub, GitLab, REST APIs, Bootstrap, LeanIX
Security	AI security, voice-assistant security, privacy, IoT security, policy-compliance analysis, threat detection, secure systems administration

Education

Clemson University
Ph.D. in Computer Science

2019–Present
Dissertation Defense Pending

Georgia Southern University <i>M.S. in Computer & Information Science</i>	2017–2019
Armstrong State University <i>B.S. in Computer Science</i>	2013–2017

Selected Publications & Research Impact

- Author or coauthor of 20+ peer-reviewed publications spanning AI security, machine learning, IoT, predictive analytics, privacy-preserving authentication, financial modeling, and data mining.
- *SkillDetective: Automated Policy-Violation Detection of Voice Assistant Applications in the Wild*, USENIX Security, 2022.
- *Dangerous Skills Got Certified: Measuring the Trustworthiness of Skill Certification in Voice Personal Assistant Platforms*, ACM CCS, 2020.
- Additional publications in IEEE, ICCS, CSCI, HONET, RFID, and related venues.

Professional Service & Recognition

Review Service	USENIX Security, IEEE INFOCOM, IEEE GLOBECOM, ACSAC, IEEE IPCCC, IEEE ESSCA, and ICCN.
Honors	Talford Family Annual Fellowship in Computing; Bhatia Foundation Scholarship in Computer Science; Dixie Crystal Scholarship.
Memberships	Upsilon Pi Epsilon, Phi Kappa Phi, IEEE, IEEE Computer Society, IEEE Young Professionals, IEEE Power and Energy Society, IEEE Council on RFID.